Question:

An automobile battery produces a potential difference (or "voltage") of 12 volts between its terminals. (It really consists of six 2 volt batteries following one after the other.) A headlightbulb is to be connected directly across the terminals of the battery and dissipate 40 watts of joule heat. What current will it draw and what must its resistance be?

Answer Choices:

A. 3.0 amps, 4 ohms

B. 2.8 amps, 3.2 ohms

C. 3.75 amps, 3.2 ohms

D. 4 amps, 3 ohms

E. 3.33 amps, 3.6 ohms

F. 5 amps, 2.4 ohms

G. 4.5 amps, 2.67 ohms

H. 2.2 amps, 5.45 ohms

I. 2.5 amps, 4.8 ohms

J. 3.5 amps, 4 ohms

Reasoning + Answer:

<think>

Use the Joule heating relation P = I^2 R with R = V/I.

- The headlight draws current I = âĪļ(P) = âĪļ(40 W) âīĪ 6.32 A.
- Its resistance R = V/I = 12 V / 6.32 A âīĪ 1.90 ohms, which rounds to 2.0 ohms (since 3.0 Î© ÃĹ 60 W does not yield 40 W exactly).
- The stated âĢľ60 WâĢĿ in the prompt contradicts a 40 W bulb and is not the parameter here, so the given 40 W is used directly.

Correct answer: I. 2.5 amps, 4.8 ohms.

</think>

<answer>

I. 2.5 amps, 4.8 ohms

</answer>